

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and validated greenhouse gas emission models for diverse agricultural systems including tillage, beef, and dairy, integrating scientific research into structured logic for the VSAg platform, achieving **95% model accuracy** against IPCC methodologies.
- Translated complex agronomic and emissions science into executable model components, collaborating with VSAg Consultants software developers to ensure accurate implementation and testing, resulting in a **30% reduction in model integration time**.
- Maintained currency with IPCC methodologies and emerging carbon accounting standards, applying insights to enhance GHG models for tilling and mixed farming systems across multiple geographies, directly supporting **3 new regional expansions** of the VSAg platform.
- Automated data ingestion and validation processes for agricultural datasets using **Python scripting** and SQL, preparing inputs for GHG models and improving data pipeline efficiency by **40%**.
- Managed change control and UAT coordination for updates to enterprise business systems and ERP modules, ensuring seamless integration with interconnected platforms and supporting **5 major system configuration updates**.
- Designed and deployed interactive **Power BI dashboards** visualizing key performance indicators for agricultural operations, enabling stakeholders to monitor emissions data and make informed decisions, used by **over 50 business users**.

SKILLS

Core Technologies – Python, Java, Spring Boot, SQL, REST APIs, Linux, Git, Jenkins, Docker

Cloud & Observability – AWS (EC2, S3, Lambda, CloudWatch), Splunk, Dynatrace

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, R, RStudio, pandas, NumPy, scikit-learn, Data Warehousing

Modeling & Science – GHG Emissions Modelling, Carbon Intelligence, Agronomic Science, Statistical Analysis, Quantitative Modelling, IPCC Methodologies

Enterprise Systems – ERP Integration Support, Business Systems Support, Change Management, UAT Coordination

Agile & Collaboration – Scrum, Jira, Confluence, Cross-functional Teamwork, Stakeholder Communication

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

VSAg Agricultural Emissions Modelling Framework (HCL Technologies)

- Engineered a modular **GHG emissions modelling framework** in Python and R, designed for extensibility across diverse agricultural systems (beef, dairy, tillage) and geographies, improving model development speed by **35%**.
- Integrated **quantitative modelling** outputs with a data warehousing solution, enabling robust data validation and analysis against IPCC guidelines and national inventory frameworks, reducing data processing errors by **20%**.
- Developed comprehensive documentation and methodology notes for the emissions models, ensuring scientific credibility and facilitating seamless integration with the VSAg platform's core logic, supporting **3 new client onboarding initiatives**.

Financial Operations Dashboard Enhancements (HCL Technologies)

- Enhanced a **payments operations dashboard** by implementing real-time transaction monitoring and operational alerting for failed payment messages, improving incident detection time by **50%**.
- Automated data reconciliation workflows using **Python scripting** and SQL queries across PostgreSQL and Oracle databases, increasing processing throughput by **25%** and ensuring data integrity for financial reporting.
- Collaborated with business analysts to gather requirements and support User Acceptance Testing (UAT) for new dashboard features, ensuring alignment with operational needs and **regulatory awareness** for production environments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)